



Cover story | CHALLENGES OF CHANGE

be about 100 years from now. An invention company has now come up with the idea of spraying sulphur dioxide into the stratosphere. Wouldn't it be a simple solution if you could introduce a chemical to stop global warming?

Take another: Almost 20 years ago, John Szilagyi, an Internal Revenue Service researcher with the US noticed that divorced couples were double-claiming their children for tax benefits. A simple idea—writing the child's social security number instead of just the name on the tax form—removed seven million children from the tax rolls in 1897. Taxpayers were just making up names because the incentive to cheat was so easy. A straightforward solution that netted in \$4 billion in one year.

I am in favour of cheap, simple solutions. But one last word: If you want ideas that can change the world to be simple and cheap, people need to be incentivised—for more than a fraction of the gain that they produce.

"We need an idea to create a green zone"

GERALD ALLAN PRESIDENT,
CRITERIA ARCHITECTS INC.

The five great challenges facing the world today come from the intersection of human needs and the natural environment. First up, the fear factor—hostility to unfamiliar people, ideas, and places—is genetic. And that leads to challenge two: Control and dominance over one's surroundings for a sense of safety. Next is the human need for culture, learning and communication. Challenge four: The need for work and worth. Finally, supply and demand, or human economics placed on top of bioeconomics. We have always used future generations and the biosphere to subsidise our actions. The fallout? Pollution, health problems, water shortage, depleted resources, endangered species and global warming. The price will continue to rise until we pay this debt. We need to rethink our strategies of state management.

To understand our challenges, let us take a look at the intersection between the environment and human needs. The rainforest is nature's most complex symbiotic environment. The city is man's most complex, interdependent creation. Fifty per cent of all species live in rainforests. Fifty per cent of all of us live in cities. Rainforests occupy 3 per cent of the land mass



(FROM LEFT) DUBNER, ALLAN, INFOSYS CO-CHAIRMAN AND SESSION CHAIRPERSON NANDAN NILEKANI AND WANG

of the planets. Cities occupy about 3 per cent of all the mass on the planets. Given the same inputs of sun, rain and soil, rainforests must maintain themselves. Cities import 75 per cent of their resources to sustain the urban environment. The output of rainforests is rich, has vast biodiversity, no wastes, produces actions that reduce carbon dioxide for the planet. Cities produce jobs, technology, creativity and culture, but also sewage, contaminated water and garbage that pollute the planet. Rainforests use the concept called local sustainability. Cities are a technological trap. If any part of the infrastructure fails—food, electricity, garbage removal, fresh water contamination—the environment becomes uninhabitable.

The great ideas of change will have to address the next evolution of working with the biosphere: A green transformer zone that will change pollution to fresh air, conserve land and control pollution. With the Internet, electronic communication and social networking, we now have the opportunity to create a global brain. But the challenge of change is more about leadership and the ability to choose wisely and make peace for the planet.